

Quiz 1 — Solutions

M 375T Fall 2026 — Q1, Q2: 3 pts each; Q3: 4 pts.

1. $(3, 2)$ and $[6 \ 9]$ (columns $0+2+4$ and $1+3+5$). *Shape right but sum along the wrong axis ($[1 \ 5 \ 9]$): 2 pts.* [3 pts]
2. $f(x) = \frac{1}{4}$ on $[0, 4]$ (0 elsewhere); $\mathbb{E}[X] = \int_0^4 \frac{x}{4} dx = 2$; $\mathbb{E}[X^2] = \int_0^4 \frac{x^2}{4} dx = \frac{16}{3}$, so $\text{Var}(X) = \frac{16}{3} - 4 = \frac{4}{3} \approx 1.333$; $\mathbb{P}(X \in [1, 2]) = \int_1^2 \frac{1}{4} dx = \frac{1}{4}$. *1 pt for density + mean, 1 pt for the variance (the integral must appear — this is Practice 2 B2 verbatim), 1 pt for the probability.* [3 pts]
3. $Av = (4, 4, 4)^\top = 4v$, so v is an eigenvector with $\lambda = 4$; $Aw = (1, -1, 0)^\top = 1 \cdot w$, so w is an eigenvector with $\lambda = 1$; $Au = (2, 1, 1)^\top$, not a multiple of u , so u is **not** an eigenvector. *Scoring: 1.5 pts each for v and w (0.5 for “eigenvector”, 1 for the eigenvalue, shown via $Av = \lambda v$); 1 pt for ruling out u with the computation. The whole point is the definition $Av = \lambda v$: multiply and compare.* [4 pts]